Amirhossein Eskorouchi

Ph.D. Candidate, Graduate Research Assistant
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Education

Ph.D. in Industrial and Systems Engineering **Anticipated Completion: May 2027**
Mississippi State University (GPA: 4.00) Mississippi, USA
Dissertation title: *Evidence-Aligned Constrained Learning for Reliable Visual AI Applications*
Advisor: Dr. Haifeng Wang, Co-advisor: Dr. Mohammad Marufuzzaman

M.S. in Industrial Engineering **September 2021**
Iran University of Science and Technology (GPA: 3.52) Tehran, Iran
Thesis title: *Construction of an Optimal Portfolio under Uncertainty Based on Evidence Theory*
Advisor: Dr. Emran Mohammadi

B.S. in Industrial Engineering **September 2018**
Garmsar University (GPA: 3.00) Semnan, Iran
Thesis title: *Feasibility Study of a Hospital Construction Project Using Simulation under Uncertainty*
Advisor: Amin Moshtagh

Research Interests

- Reliable and decision-aligned artificial intelligence through knowledge-informed, evidence-constrained, and optimization-integrated machine learning
- Data-efficient, transferable, and adaptive learning under limited supervision, data heterogeneity, and distributional change using self-supervised learning, domain adaptation, neural architecture search, and optimal transport
- Computer vision and structured and multimodal data analysis for image classification, object detection, semantic and instance segmentation, predictive modeling, and interpretable decision support
- Intelligent decision-support systems integrating machine learning with edge computing, multimodal sensing, IoT platforms, and robotic systems

Application Areas:
Medical image analysis and clinical decision support; intelligent manufacturing and quality monitoring; smart agriculture and biomass processing.

Research Experience

Graduate Research Assistant, Mississippi State University **January 2024 – Present**
Graduate Research Assistant, Iran University of Science and Technology **Sep 2021 – June 2023**

Responsibilities:

- Conduct scholarly activities

- Co-mentor students in their scholarly pursuits
- Develop and evaluate machine learning models, research datasets, and AI-enabled prototypes
- Collaborate with interdisciplinary research teams
- Write and publish scholarly articles in journals and present research findings at conferences

List of Projects & Topics:

1. **Project:** Design and Field Validation of a Portable Edge-AI System for Forest Biomass Quality Assessment
   Sponsor: University of Tennessee, Knoxville No. A26-0917
   Research Period: April 2026 – Present

2. **Project:** Image-Based Assessment of Wood Chip Attributes
   Sponsor: USDA–NIFA Nos. 2020-67019-30772 and 2022-67022-37861
   Research Period: January 2024 – Present

3. **Project:** Image-Guided Automated Extracapsular Extension Detection in Head and Neck Cancer
   Sponsor: NIH No. 1R03DE032766-01
   Research Period: January 2024 – Present

4. **Topic:** Validating an Artificial Intelligence Powered Multimodal Sensing and Data Collection Platform for Plant Health Management
   Sponsor: System Intelligence and Optimization Lab (May 2026)

5. **Topic:** Transferable Neural Architecture Search for Medical Image Classification
   Sponsor: System Intelligence and Optimization Lab (March 2026)

6. **Topic:** Machine Learning-Based Polysomnography Data Analysis for ADHD Diagnosis
   Sponsor: System Intelligence and Optimization Lab (May 2025)

7. **Topic:** Dynamic Building Risk Analysis and Decision-Support Tool
   Sponsor: System Intelligence and Optimization Lab (May 2024)

Journal Publications

Google Scholar Profile: Citations 55, h-index 4, i10-index 3 (since 2021; as of August 6, 2026).

Published/In Press/Accepted

J1. Vo, A., He, L., **Eskorouchi, A.**, & Wang, H. (2026). "Understanding Knowledge Transferability in Differentiable Architecture Search for Skin Lesion Diagnosis." *IISE Transactions on Healthcare Systems Engineering*, 16(1), 71–90. doi:10.1080/24725579.2026.2622416.

J2. Mohammadi, E., Karimi, M. H., **Eskorouchi, A.**, & Ghanbari, H. (2025). "Dempster–Shafer-Informed Levelized Cost of Energy: A Robust Approach for Energy Portfolio Optimization under Deep Uncertainty." *Results in Engineering*, 28, 107756. doi:10.1016/j.rineng.2025.107756.

J3. **Eskorouchi, A.**, Ghanbari, H., & Mohammadi, E. (2023). "Exploring the Evolution of Robust Portfolio Optimization: A Scientometric Analysis." *Iranian Journal of Accounting, Auditing and Finance*. doi:10.22067/ijaaf.2024.44518.1402.

J4. Ghanbari, H., Larni Fooeik, A. M., **Eskorouchi, A.**, & Mohammadi, E. (2022). "Investigating the Effect of US Dollar, Gold and Oil Prices on the Stock Market." *Journal of Future Sustainability*, 2(3), 97–104. doi:10.5267/j.jfs.2022.9.009.

J5. Karimi, M. H., Mohammadi, E., Jafari, H., Ghaeli, M. R., & **Eskorouchi, A.** (2022). "A Robust Linear Model for the Maximum Expected Coverage Location Problem Considering the Relative Coverage." *Journal of Future Sustainability*, 2(2), 39–48. doi:10.5267/j.jfs.2022.9.002.

J6. **Eskorouchi, A.**, Mohammadi, E., & Sadjadi, S. J. (2022). "Robust Portfolio Optimization Based on Evidence Theory." *Scientia Iranica*. doi:10.24200/SCI.2022.59575.6330.

J7. **Eskorouchi, A.**, & Mohammadi, E. (2022). "Uncertain Portfolio Optimization Based on Dempster–Shafer Theory." *Management Science Letters*, 12(3), 207–214. doi:10.5267/j.msl.2022.1.001.

Under Review

J8. **Eskorouchi, A.**, Street, J. T., Wang, H., & Marufuzzaman, M. "UOT-DETR: Distributionally Constrained Visual Quality Measurement with Unbalanced Optimal Transport for Wood-Chip Size Estimation," submitted to: *Knowledge-based Systems* (1st Review).

J9. **Eskorouchi, A.**, Rahman, A., Street, J., Marufuzzaman, M., & Wang, H. "WoodChip-MC and WoodChip-Detection: Public Image Datasets for Vision-Based Wood Chip Quality Evaluation," submitted to: *Scientific Data* (1st Review).

In Preparation

- **Eskorouchi, A.**, Duggar, W. N., & Wang, H. "AnatoECE: Knowledge-Informed Anatomy-Constrained Learning with Self-Supervised Pretraining for Extranodal Extension Assessment under Limited Labeled Data." Manuscript in preparation for submission to *Medical Image Analysis*.
- **Eskorouchi, A.**, Kang, D. E., Wang, H., & Marufuzzaman, M. "A Cost-Effective Prototype for Multimodal IoT-Based Agricultural Monitoring."
- **Eskorouchi, A.**, Wang, H., & Marufuzzaman, M. "Evidence-Aligned Constrained Learning for Reliable Visual AI Applications."

Public Datasets

D1. **Eskorouchi, A.**, Street, J. T., Wooten, J., Marufuzzaman, M., & Wang, H. "WoodChip-Detection: A Public Dataset for Dense Wood Chip Detection and Instance Segmentation." *Zenodo*, Version 1.0, 2026. doi:10.5281/zenodo.18392693.

Conference Publications

Published in Proceedings

C1. Wang, Y., Pias, A. S., **Eskorouchi, A.**, Duggar, W. N., Wang, H., & Fan, L.-W. "DA-GMGENet: A Domain Adaptation Framework for 3D CT-Based ECE Identification in Head and Neck Cancer," in *Proceedings of the 2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI 2026)*, London, UK, 2026, pp. 1–4. doi:10.1109/ISBI61048.2026.11515601.

C2. **Eskorouchi, A.**, Bhattarai, P., Rahman, A., Marufuzzaman, M., Street, J. T., & Wang, H. "An Edge Artificial Intelligence System for Wood Chip Quality Evaluation," in *Proceedings of the IISE Annual Conference & Expo 2026*, Arlington, TX, 2026, pp. 1–6.

C3. **Eskorouchi, A.**, Wang, H., Lee, J. W., Nayak, V. H., Ojeda, N. B., & Fan, L.-W. "Machine Learning-Based Polysomnography Data Analysis for ADHD Diagnosis: A Focus on Sleep Stage-Based Biomarkers," in *Proceedings of the 2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI 2025)*, Houston, TX, 2025, pp. 1–4. doi:10.1109/ISBI60581.2025.10981031.

C4. **Eskorouchi, A.**, Rahman, A., Street, J. T., Marufuzzaman, M., & Wang, H. "Enhanced DETR-Based Framework for Automated Wood Chip Size Distribution Estimation in High Volume Biomass Manufacturing," in *Flexible Automation and Intelligent Manufacturing: The Future of Automation and Manufacturing: Intelligence, Agility, and Sustainability*, Springer, Cham, 2025, pp. 540–548. doi:10.1007/978-3-032-05610-8_53.

C5. **Eskorouchi, A.**, Duggar, W. N., & Wang, H. "Knowledge-Informed Automated Extracapsular Extension Detection in Head and Neck Cancer," in *Proceedings of the IISE Annual Conference & Expo 2025*, Atlanta, GA, 2025, pp. 1–6.

C6. **Eskorouchi, A.**, Rahman, A., Marufuzzaman, M., Street, J. T., & Wang, H. "Automated Wood Chip Size Estimation with YOLO-Based Object Detection," in *Proceedings of the IISE Annual Conference & Expo 2025*, Atlanta, GA, 2025, pp. 1–6.

C7. Rahman, A., **Eskorouchi, A.**, Street, J. T., Wooten, J., Marufuzzaman, M., & Wang, H. "Wood Chip Moisture Content Assessment Using Infrared Image-Based Machine Learning," in *Proceedings of the ASABE International Meeting 2024*, Anaheim, CA, July 28–31, 2024, pp. 1–12.

Peer-Reviewed Abstracts

(underlining denotes the presenter)

A1. **Eskorouchi, A.**, Wang, H., & Marufuzzaman, M. "UOT-DETR: An End-to-End Unbalanced Optimal Transport Transformer for Dense Small-Object Detection and Size Estimation in Wood-Chip Manufacturing," INFORMS Annual Meeting 2026, San Francisco, CA, November 1–4, 2026.

A2. **Eskorouchi, A.**, Yuan, L., Duggar, W. N., & Wang, H. "A Self-Supervised Transformer Framework for Interpretable Extranodal Extension Detection In Head and Neck Cancer," 2026 Joint AAPM–COMP Meeting, Vancouver, British Columbia, Canada, July 19–22, 2026.

A3. **Eskorouchi, A.**, Duggar, W. N., & Wang, H. "Knowledge-Informed Deep Learning for Accurate and Interpretable Extracapsular Extension Detection in Head and Neck Squamous Cell

Carcinoma," American Association of Physicists in Medicine 67th Annual Meeting & Exhibition (AAPM 2025), Washington, DC, July 27–30, 2025; abstract published in *Medical Physics*, 52(10), e70059111, October 2025.

A4. **Eskorouchi, A.**, Duggar, W. N., Nittala, M. R., Purdom, K. E., & Wang, H. "Impact of Human Papillomavirus Status on Artificial Intelligence-Based Extracapsular Extension Detection," 67th American Society for Radiation Oncology Annual Meeting (ASTRO 2025), San Francisco, CA, September 27–October 1, 2025.

A5. **Eskorouchi, A.**, Lee, J. W., Fan, L.-W., Wang, H., Nayak, V. H., Ojeda, N. B., Chen, Z., & Nadorff, M. "Machine Learning-Based Polysomnography Analysis for ADHD: Sleep Stage-Derived Biomarkers for Improved Diagnosis," 40th Southern Biomedical Engineering Conference (SBEC), Louisiana State University Health Shreveport, Shreveport, LA, September 13–15, 2024.

A6. **Eskorouchi, A.**, Duggar, W. N., & Wang, H. "A Gradient Mapping Guided Explainable Deep Neural Network for Extracapsular Extension Identification," 40th Southern Biomedical Engineering Conference (SBEC), Louisiana State University Health Shreveport, Shreveport, LA, September 13–15, 2024.

Poster Presentations

(underlining denotes the presenter)

P1. **Eskorouchi, A.** & Wang, H. "Knowledge-Informed Constrained Learning Under Limited Supervision for Head and Neck Cancer Evaluation," IISE Annual Conference & Expo 2026, Arlington, TX, 2026.

P2. **Eskorouchi, A.**, Rahman, A., Street, J. T., Marufuzzaman, M., & Wang, H. "Enhanced DETR-Based Framework for Automated Wood Chip Size Distribution Estimation in High Volume Biomass Manufacturing," IISE Annual Conference & Expo 2025, Atlanta, GA, 2025.

P3. **Eskorouchi, A.**, Wang, H., Lee, J. W., Nayak, V. H., Ojeda, N. B., & Fan, L.-W. "Machine Learning-Based Polysomnography Data Analysis for ADHD Diagnosis: A Focus on Sleep Stage-Based Biomarkers," 2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI 2025), Houston, TX, April 14–17, 2025.

Software & Prototypes

- **PlantLife365: Multimodal IoT-Based Agricultural Monitoring Platform**
  Co-developed a multisensor IoT device and web application for real-time agricultural monitoring, historical visualization, configurable alerts, data export, and machine-learning-based predictive analysis.

- **Wood-Chip Monitor: Edge-AI Quality Evaluation System**
  Designed a portable edge-AI prototype and browser-based dashboard for real-time wood-chip detection, size-distribution measurement, moisture classification, quality alerts, and event reporting.

- **Extracapsular Extension Detection Web Application**
  Developing a web interface for CT-based extracapsular extension detection, allowing users to upload medical images and obtain model predictions with visual interpretability support.

- **Dynamic Building Risk Analysis Web Application**
  Built an interactive decision-support application for multi-hazard building risk assessment, damage estimation, annual-risk calculation, net-present-value analysis, and map-based visualization.

Teaching Interests

My teaching interests span three interconnected areas: artificial intelligence and data analytics, optimization and decision sciences, and industrial systems and management. I am enthusiastic about teaching undergraduate and graduate courses in machine learning, deep learning, computer vision, data analytics, operations research, and linear and nonlinear programming. I am also interested in teaching core industrial engineering courses, including engineering statistics, engineering economy, financial engineering and risk management, operations and supply chain management, and information systems, as well as other courses aligned with departmental needs.

Teaching & Mentoring Experience

Teaching Activities

Guest Lectures and Invited Instruction

- **Invited Instructor**, USDA REEU *AI-Enabled Industrial Solutions for Forest Products* Summer Program, Mississippi State University **June 26, 2026**
  "Deep Learning with PyTorch."

- **Guest Lecturer**, Graduate Course IE 8703: Optimization in Deep Learning, Department of Industrial and Systems Engineering, Mississippi State University **February 19, 2025**
  "Tutorial on LaTeX and Overleaf."

Teaching Assistantship

- **Teaching Assistant**, Iran University of Science and Technology **February 2021 – June 2022**
  Courses: Principles of Financial Engineering; Financial Risk Management
  - Supported course instruction and prepared instructional materials
  - Assisted students with quantitative concepts and problem-solving activities
  - Graded assignments and examinations and provided feedback
  - Assisted with course administration and student performance evaluation

Student Co-Mentoring

- **Abhro Shome Pias** (Ph.D. Student, Spring 2025 – Present)
  Project: Domain adaptation for 3D CT-based extracapsular extension identification in head and neck cancer
  Outcome: Conference paper published in the *Proceedings of IEEE ISBI 2026*

- **Anh Vo** (Master's Student, August 2025 – December 2025)
  Project: Differentiable neural architecture search and knowledge transferability for skin-lesion diagnosis
  Outcome: Journal article published in *IISE Transactions on Healthcare Systems Engineering*, 2026

- **Mohammad Hossein Karimi** (Master's Student, August 2022 – August 2023)
  Project: Dempster–Shafer-informed energy portfolio optimization under deep uncertainty
  Outcome: Journal article published in *Results in Engineering*, 2025

Undergraduate Research Mentoring **Summer 2026**
USDA-supported Research and Extension Experiences for Undergraduates (REEU) Program: *AI-Enabled Industrial Solutions for Forest Products (AI2F)*

- **Joseph (Trey) Dooley**
  Research Project: Comparative Evaluation of Transfer Learning Models for Bark Beetle Gallery Classification

- **Dustin T. Gavins**
  Research Project: A Deep Learning Application for Pine Fungal Disease Classification

- **Cole Thibodeaux**
  Research Project: Development of a Portable AI-Powered Field Device for Forest Monitoring Applications

- **Kayla A. Pigott**
  Research Project: The AI of the Tiger: AI-Based Wood Pattern Classification of Tiger-Stripe Wood Veneers

- **Mandy Dreznick**
  Research Project: Determining Wood-Chip Moisture Content Using Deep Adaptive Boosting

Invited Presentation

- **Invited Speaker**, USDA REEU *AI-Enabled Industrial Solutions for Forest Products* Summer Program, Mississippi State University, 2025. "Automated Wood-Chip Size Distribution Estimation in High-Volume Biomass Manufacturing."

Academic & Professional Service

- **Peer-Review Service**
  - **Ad Hoc Reviewer for Academic Journals:** *Journal of Cleaner Production*; *Engineering Applications of Artificial Intelligence*; *Computers & Operations Research*; *Optics & Laser Technology*; and *Journal of the Mississippi Academy of Sciences*
  - **Conference Proceedings Reviewer:** *Flexible Automation and Intelligent Manufacturing 2025*, Springer Proceedings; and *IISE Annual Conference & Expo 2025*
- **Conference Leadership**
  - **Session Chair**, "Advanced AI Solutions in Health and Diagnostics," Data Analytics and Information Systems (DAIS) Track, IISE Annual Conference & Expo 2025
  - **Session Co-Chair**, "Emerging Methods in Vision-Based Decision Support," Quality Control and Reliability Engineering (QCRE) Track, IISE Annual Conference & Expo 2025
- **Professional Society Leadership**
  - **President**, INFORMS Student Chapter, Mississippi State University 2026–Present
  - **Vice President**, INFORMS Student Chapter, Mississippi State University 2025–2026
- **University & Institutional Service**
  - **Vice President**, Iranian Student Association, Mississippi State University 2025–Present
  - **President**, Iranian Student Association, Mississippi State University 2024–2025

Grants & Awards

Awards and Honors

- **Finalist**, INFORMS Data Mining Society Data Challenge, INFORMS Annual Meeting 2026
- **First Place**, Data Analytics & Information Systems (DAIS) Division Student Mobile App Competition, IISE Annual Conference & Expo, for *PlantLife365* 2026
- **Finalist**, Data Analytics & Information Systems (DAIS) Division Student Mobile App Competition, IISE Annual Conference & Expo, for *Wood-Chip Monitor* 2026
- **Third Place**, DAIS Data Analytics Competition, IISE Annual Conference & Expo 2025
- **Third Place**, Campus Vision Challenge, Mississippi State University Artificial Intelligence Club 2025

Travel Grants & Research Support

- **Conference Travel Support**, Mississippi State University 2026
- **Student Research Travel Award**, Bagley College of Engineering, Mississippi State University 2026
- **Student Research Travel Award**, Bagley College of Engineering, Mississippi State University 2025
- **Student Research Travel Award**, Bagley College of Engineering, Mississippi State University 2024

Professional Memberships

- Elected Member, Phi Kappa Phi Honor Society (top 10% of graduate students)
- Member, Institute for Operations Research and the Management Sciences (INFORMS)
- Member, Institute of Industrial and Systems Engineers (IISE)

Technical Strengths

Programming Languages	Python, R, etc.
Machine Learning & Data Mining Tools	PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, MONAI, etc.
Databases & Data Visualization	SQL, Microsoft Access, Power BI, etc.
Optimization Tools	GAMS, CPLEX, Gurobi, etc.
Simulation Tools	Arena, FlexSim, etc.
Development & Other Tools	Jupyter Notebook, PyCharm, Spyder, RStudio, LaTeX, etc.

Certifications & Professional Development

- **Equipping Future Entrepreneurs Seminar Series**, Center for Entrepreneurship and Outreach and The Graduate School, Mississippi State University Spring 2026
- **Machine Learning in Python**, 365 Data Science Spring 2021
- **SQL**, 365 Data Science Spring 2026
- **The Complete Data Visualization Course with Python, R, Tableau, and Excel**, 365 Data Science Spring 2021

References

- **Haifeng Wang, Ph.D.**, Supervisor
  Associate Professor
  Department of Industrial and Systems Engineering
  Auburn University
  Email: hfwang@auburn.edu Phone: (607) 372-9427

- **Mohammad Marufuzzaman, Ph.D.**, Advisor
  Department Head and Larry Brown Endowed Professor
  Charles R. Stephenson Endowed Professorship
  Department of Industrial and Systems Engineering
  Mississippi State University
  Email: maruf@ise.msstate.edu Phone: (662) 325-7624

- **Jason Tyler Street, Ph.D., P.E.**, Collaborator
  Interim Department Head and Associate Professor
  Department of Sustainable Bioproducts
  Mississippi State University
  Email: jason.street@msstate.edu Phone: (662) 325-5120